Session 2 — Statistics & Inference

Exercise sheet

Work through these in order: they start with hands-on calculation and build toward subtler ideas, including two classic statistical paradoxes. Show your working, and always ask what each number means, *not just how to compute it.*

Exercise 1 *Trial-by-trial firing rates (spikes per second) of a single neuron.*

The recorded rates across five trials are $6,\ 8,\ 10,\ 12,\ 14$.

(a) Compute the mean.

(b) Compute the variance (the average squared deviation from the mean) and the standard deviation.

Exercise 2 A set of test scores has mean 70 and standard deviation 8.

(a) Compute the z-score of a score of 86.

(b) If *every* score is standardised to a z-score, what are the mean and standard deviation of the resulting z-scores?

(c) If instead the scores are rescaled by $Y = 1.5X + 10$, give the new mean and standard deviation.

Exercise 3 *Two simultaneously recorded neurons; each column is one trial.*

Neuron A: $x = (1, 2, 3, 4, 5)$. Neuron B: $y = (2, 4, 5, 4, 5)$. The means are $\bar{x} = 3$ and $\bar{y} = 4$.

(a) Compute the covariance $\mathrm{Cov}(x, y)$ (divide by $n = 5$).

(b) Compute the correlation coefficient r.

(c) Interpret the sign and rough magnitude of r in one sentence.

Exercise 4 *Averaging repeated trials to suppress measurement noise (as in EEG/fMRI).*

Single-trial noise has standard deviation $\sigma = 12$ units.

(a) Compute the standard error of the mean over $n = 36$ trials.

(b) How many trials are needed to bring the standard error down to 1 unit?

(c) Explain in one sentence why this "$\sqrt{n}$" scaling means precision is expensive.

Exercise 5 A sample of $n = 64$ trials has mean reaction time 540 ms; the population standard deviation is $\sigma = 80$ ms.

(a) Compute the 95% confidence interval for the mean (use $\bar{x} \pm 1.96\,\sigma/\sqrt{n}$).

(b) Compute the 99% confidence interval (use the multiplier 2.576).

(c) Which interval is wider, and why?

Exercise 6 *Testing whether a participant performs above chance on a yes/no task.*

A subject is correct on 44 of 60 trials. Under the null hypothesis (chance = 0.5) the number correct has mean 30 and standard deviation $\sqrt{60 \cdot 0.5 \cdot 0.5} = \sqrt{15} \approx 3.87$.

(a) Compute the z-statistic.

(b) Is the result significant at the 5% level (roughly $|z| > 2$)? State your conclusion in words.

Exercise 7 *Comparing a patient group with a control group.*

Patients have mean 520 ms, controls 500 ms, and both groups have standard deviation 40 ms, with $n = 25$ per group.

(a) Compute Cohen's $d = (\text{mean}_1 - \text{mean}_2)/\text{SD}$.

(b) Compute the standard error of the difference, $\text{SD}\sqrt{1/n_1 + 1/n_2}$, and the t-statistic.

(c) Is the difference significant at 5%? Reconcile your answer with the effect size.

Exercise 8 In the same patient-vs-control setting:

(a) Define a Type I and a Type II error in plain words.

(b) Increasing the sample size most directly reduces which type of error, and why?

Exercise 9 *Predicting a neuron's firing rate y from stimulus contrast x.*

The data are $x = (0, 1, 2, 3, 4)$ and $y = (1, 1, 2, 2, 4)$, with $\bar{x} = 2$, $\bar{y} = 2$.

(a) Compute the least-squares slope b and intercept a.

(b) Predict the firing rate at contrast $x = 5$.

(c) The correlation here is $r \approx 0.90$; compute R^2 and state what it means.

Exercise 10 *A concrete look at the Central Limit Theorem.*

A single die roll has mean 3.5 and standard deviation ≈ 1.71. You average $n = 100$ independent rolls.

(a) What is the mean of the sample mean $\bar{X}$?

(b) What is its standard error?

(c) What is the approximate *shape* of the distribution of $\bar{X}$, and why?

Exercise 11 *Two independent sources of neural noise add together.*

The two sources have variances 9 and 16.

(a) Compute the variance and standard deviation of their sum.

(b) If instead the two sources were perfectly correlated, what would the standard deviation of the sum be? *(Standard deviations then add.)*

Exercise 12 A participant scores 130 on test A (mean 100, SD 15) and 60 on test B (mean 50, SD 5). On which test did they perform *relatively* better? Justify your answer with the two z-scores.

Exercise 13 **Simpson's paradox.** Two treatments are compared for recovery, split by case severity:

	Treatment A	Treatment B
Mild cases	81/87 (93%)	234/270 (87%)
Severe cases	192/263 (73%)	55/80 (69%)
Overall	273/350 (78%)	289/350 (83%)

(a) Which treatment has the higher recovery rate *within each* severity group?

(b) Which looks better *overall*?

(c) Name the paradox and explain in one sentence how it arises here.

Exercise 14 **Regression to the mean.** On a first test, the students who scored *worst* tend to improve on a retest even with no teaching in between.

(a) Explain why, in terms of luck and measurement noise.

(b) What does this imply for evaluating a "remedial programme" that enrolled only the lowest scorers, if there is no control group?

Exercise 15 A study reports a 95% confidence interval for a mean difference (patients minus controls) of $[2,\ 8]$ ms.

(a) Is the value 0 inside the interval? What does that imply about significance at the 5% level?

(b) Someone says "there is a 95% probability the true difference lies in $[2, 8]$." Explain briefly why this is the wrong reading, and give the correct one.